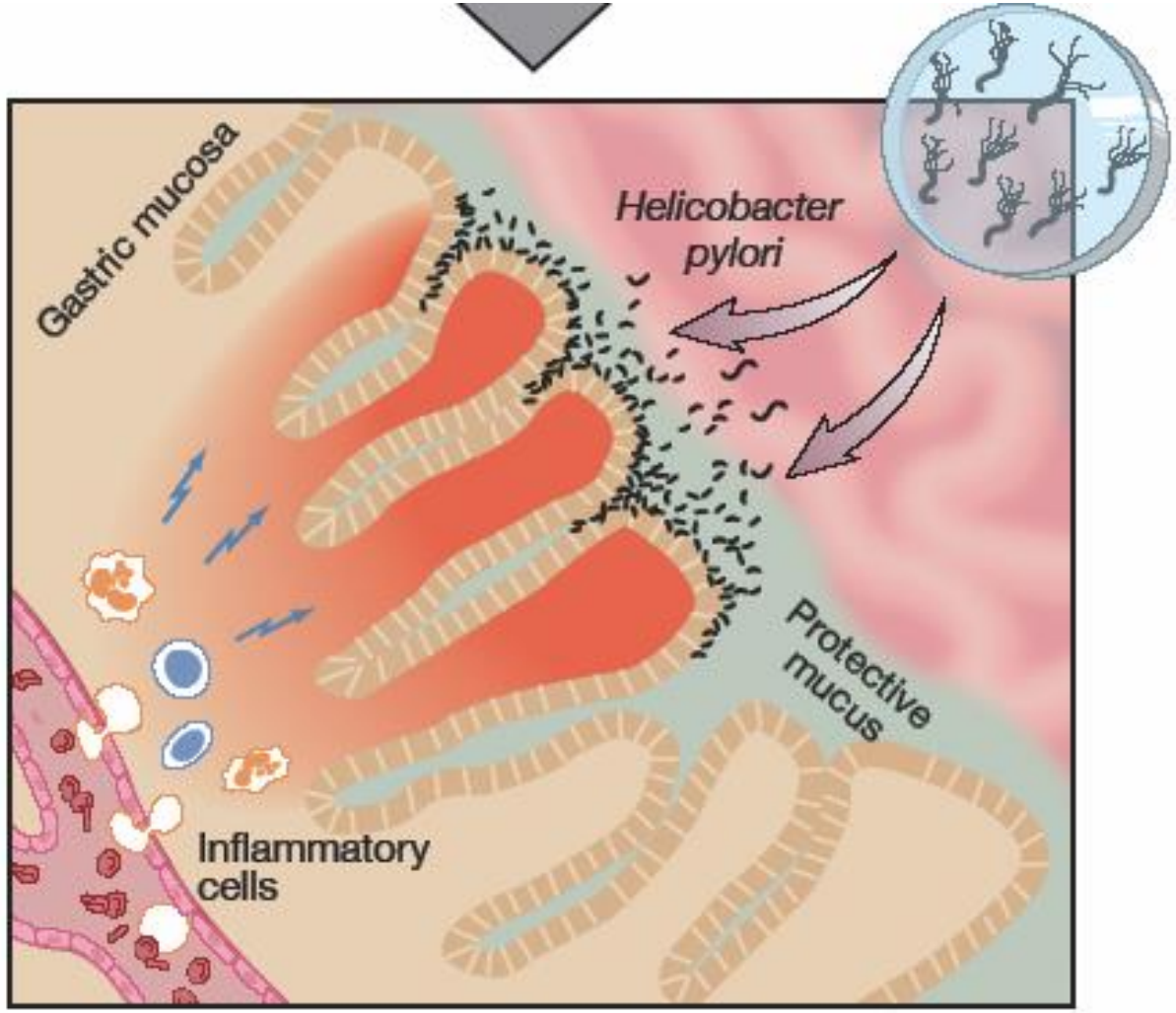
Project 1



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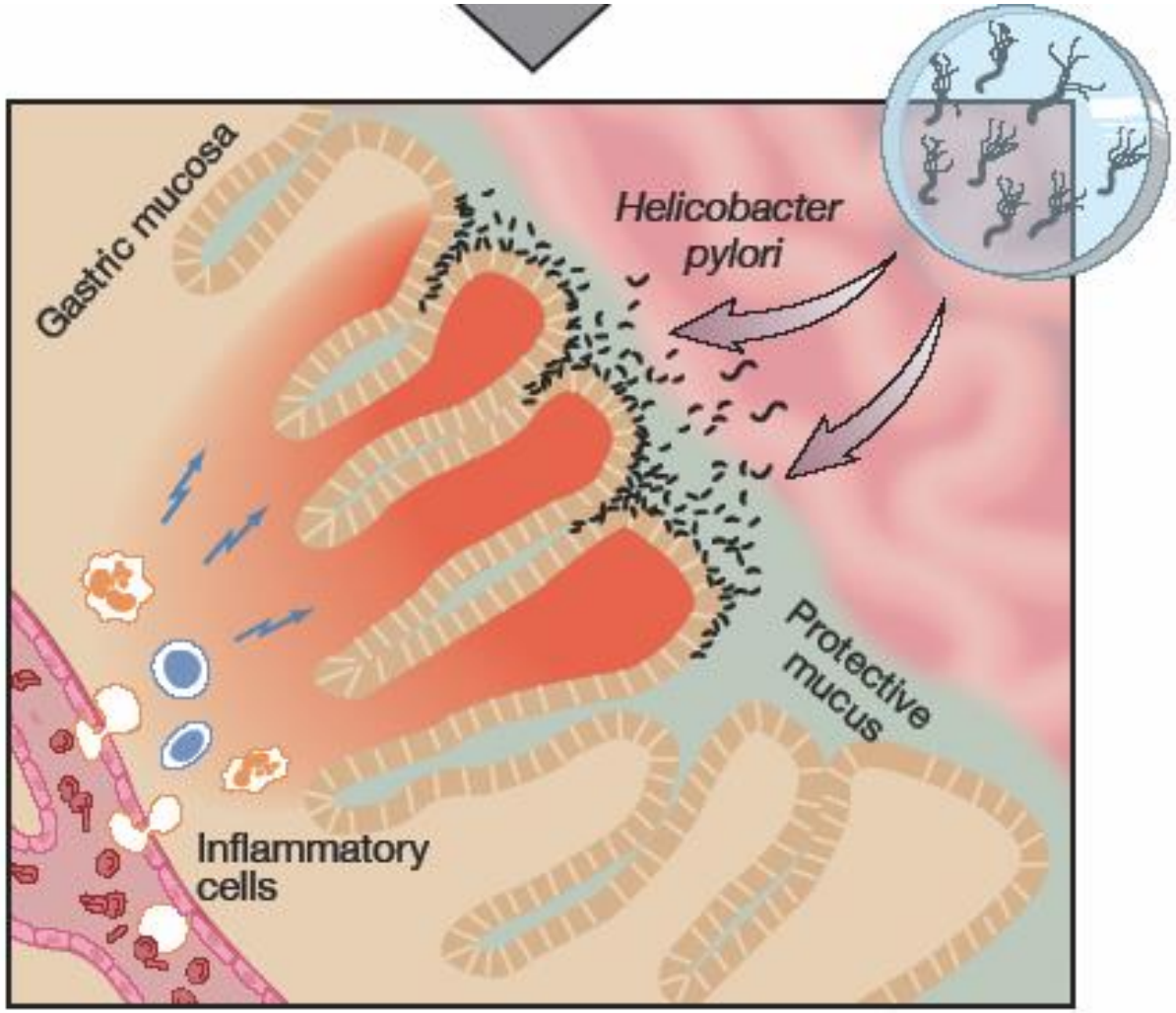
Motivation

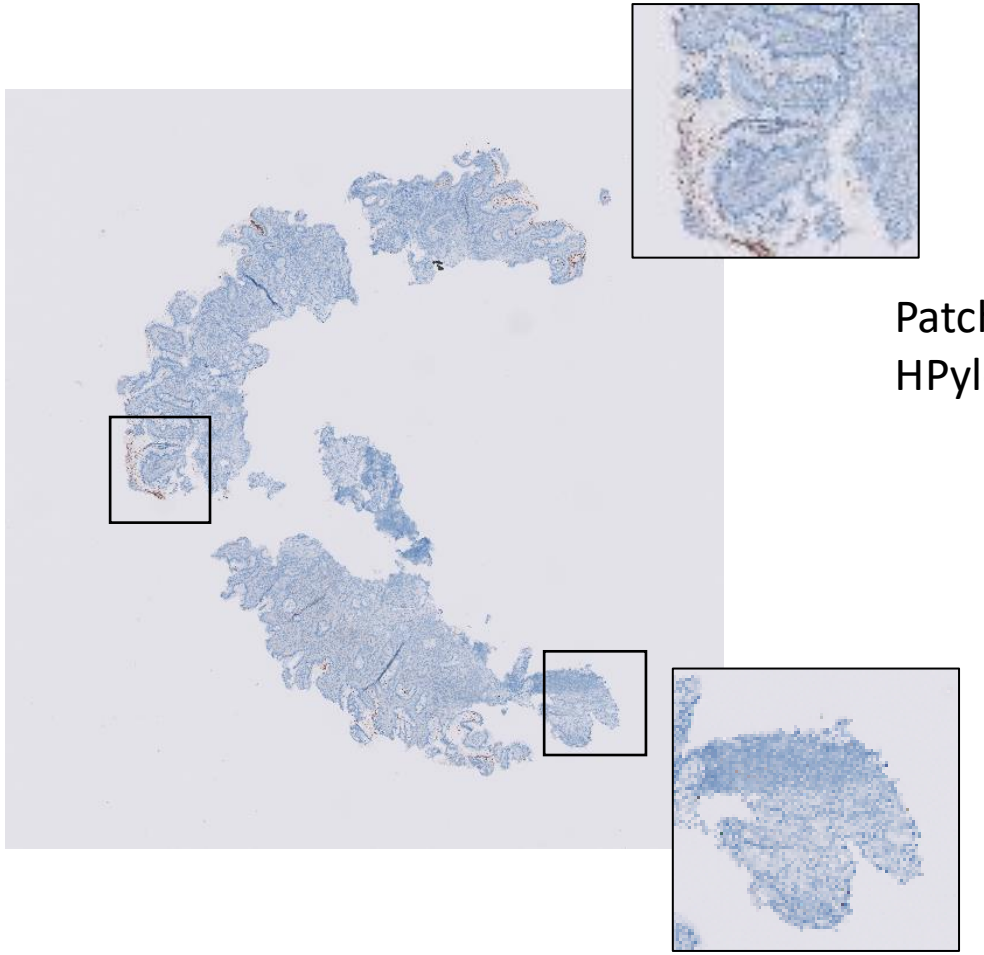
Helicobacter pylori (H.pylori) is a gastric bacterium with worldwide prevalence and is classified as a class I carcinogen.

H.pylori causes **inflammation** of the gastric mucosa (gastritis). This is often asymptomatic.



Motivation

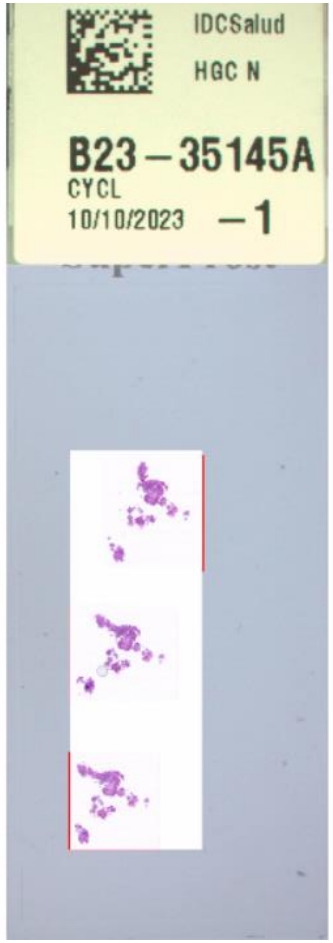
A common method for detecting H.pylori involves examining histological samples using immunohistochemical staining



Patch with HPylori

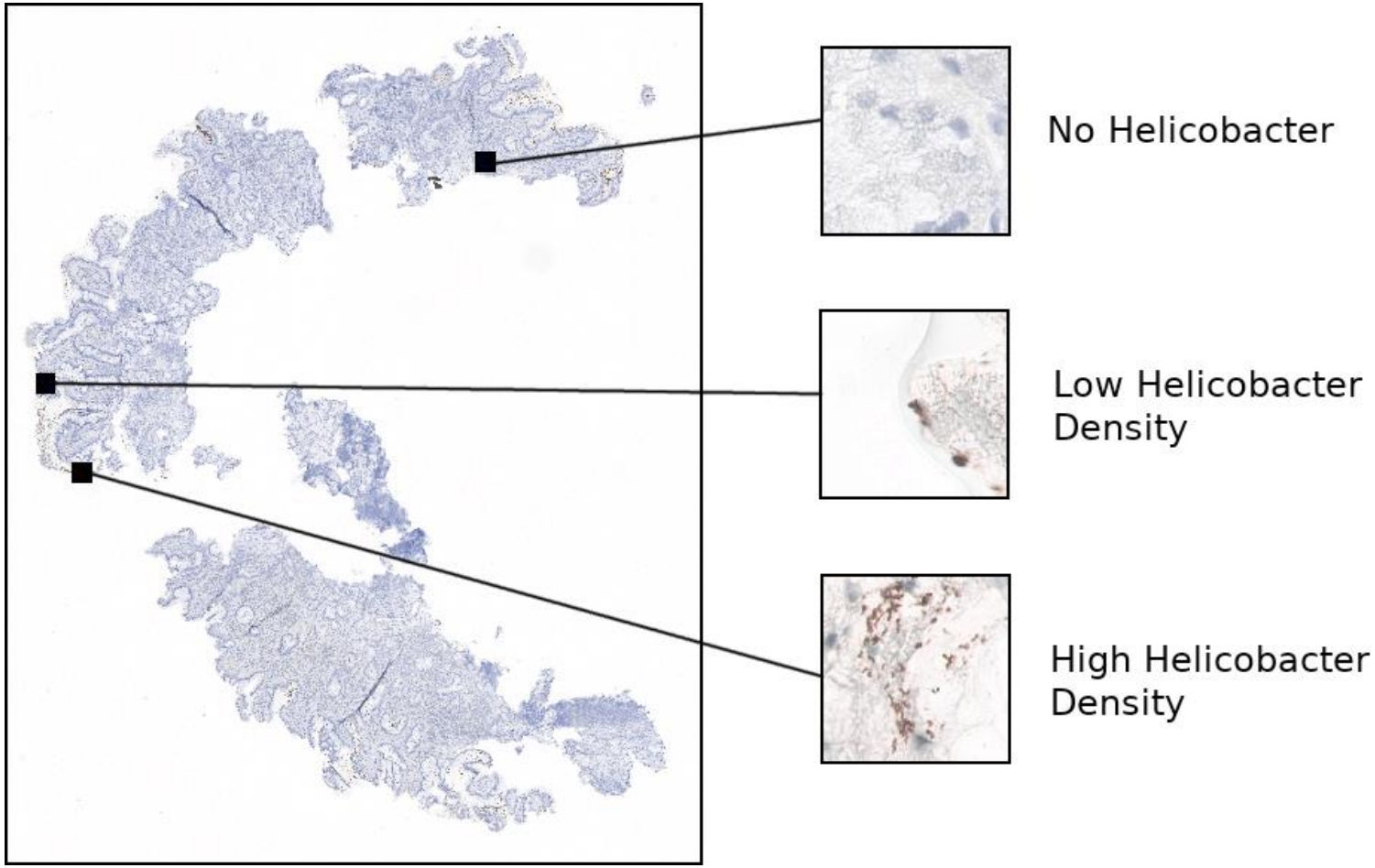
Patch without HPylori

Whole-Slide Image



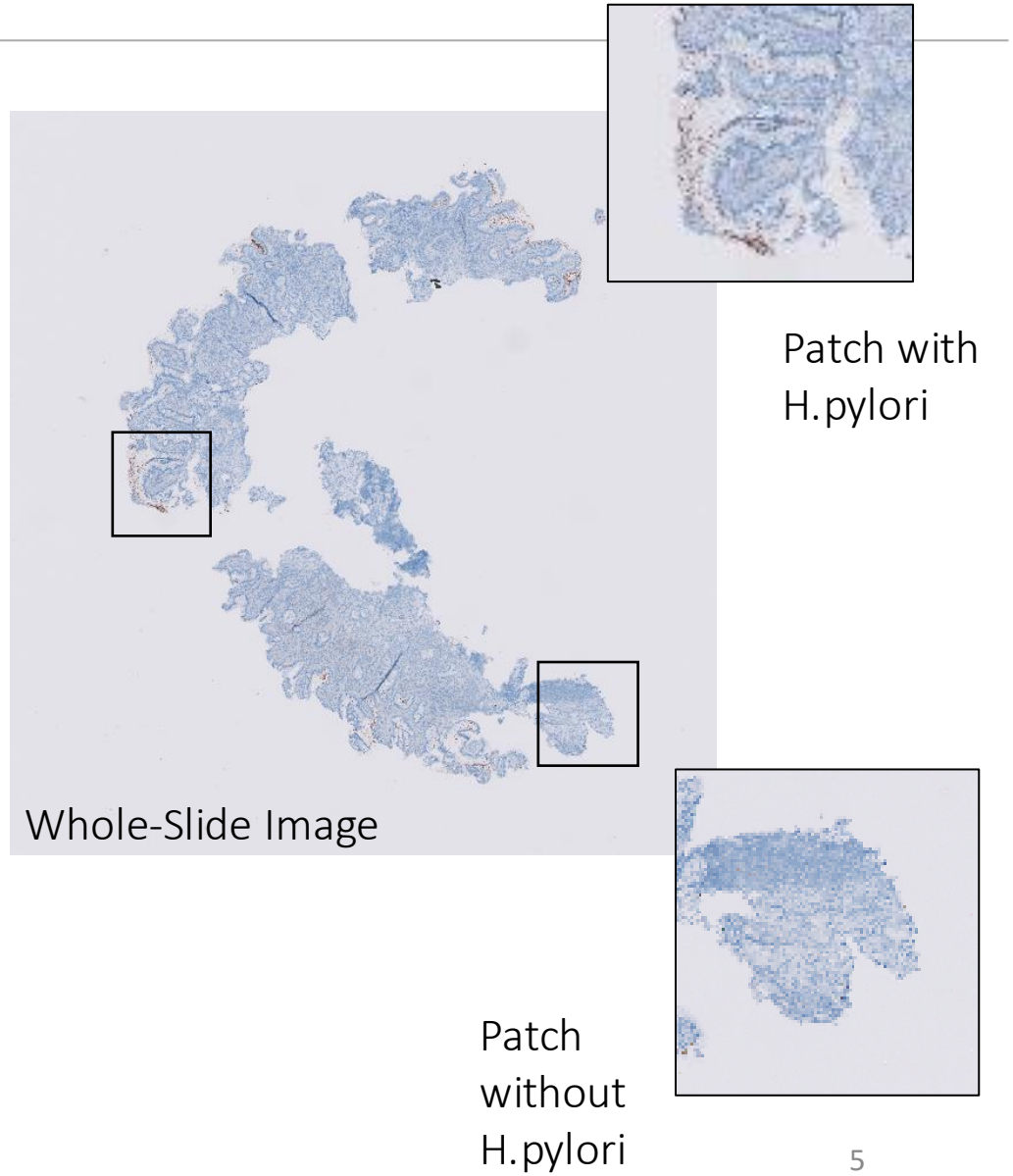
Motivation

The visual examination process is extremely time-consuming due to the large image dimensions (120.000 x 16.000 pixels) and the varying bacterial densities



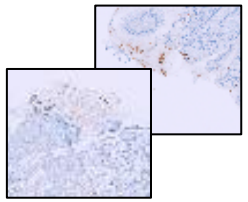
Main Objectives

- Detection of H.pylori in immunohistochemical histopathological images
- Identification of main problems in small sample size unbalanced data sets
- Detection of anomalous staining using self-supervised learning (generative models)
- Comparison to classification approach

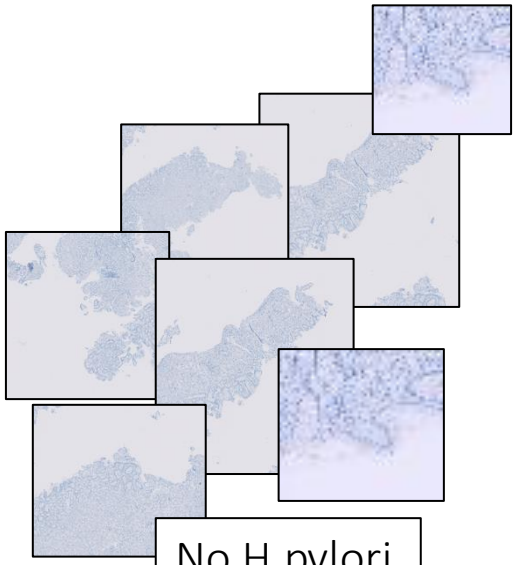


Challenges for ML-DL methods

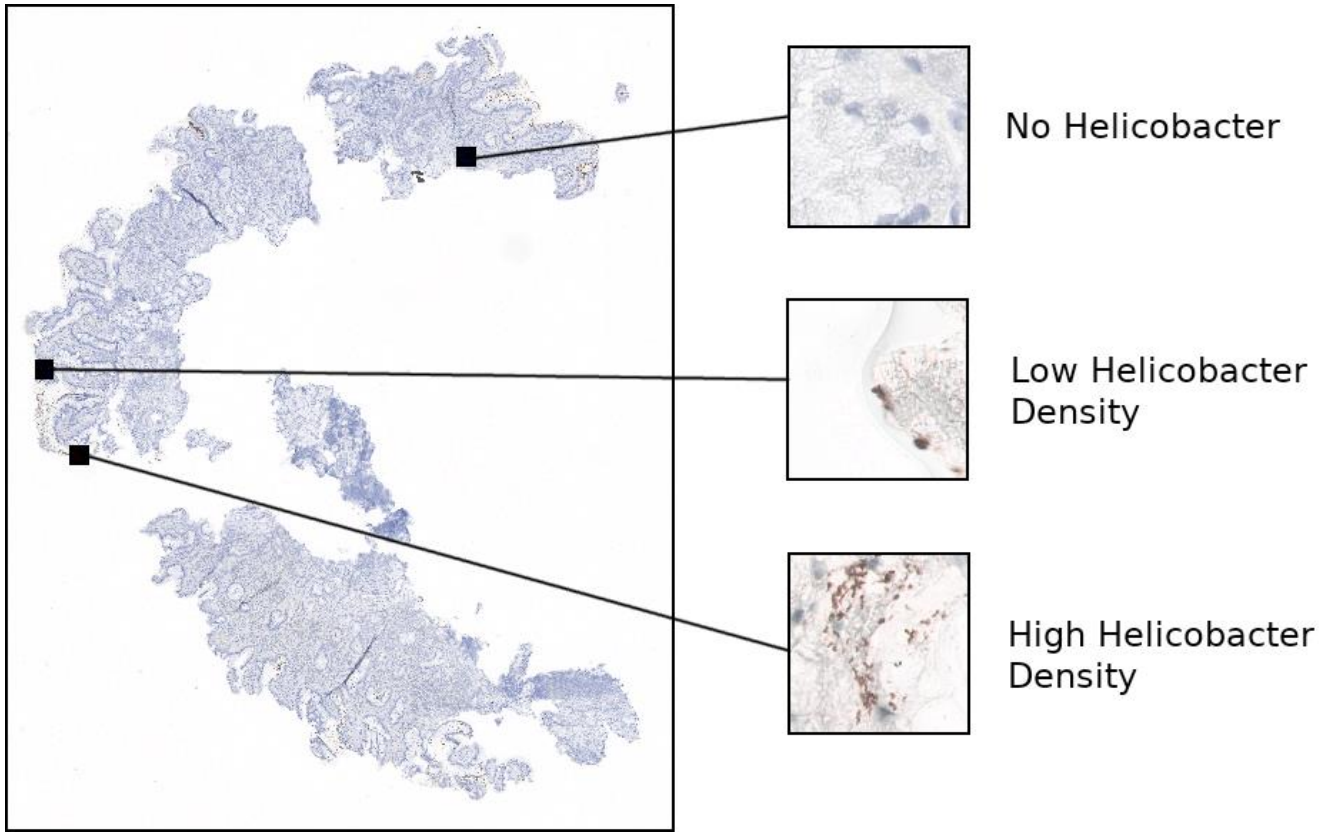
- Small sample size of WSI (245 cases) of huge size (120.000 x 16.000 pixels)
- Highly unbalance in the set of patches
- Different densities of H.pylori



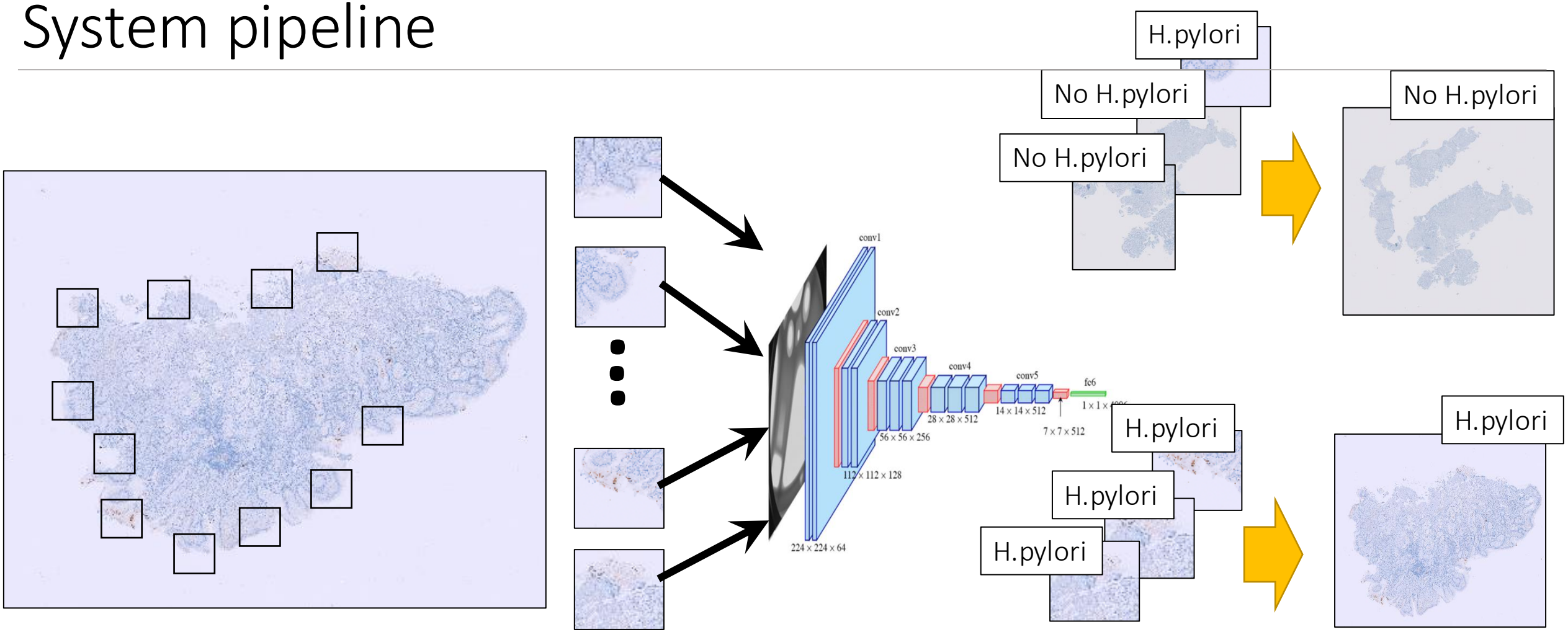
H.pylori



No H.pylori



System pipeline



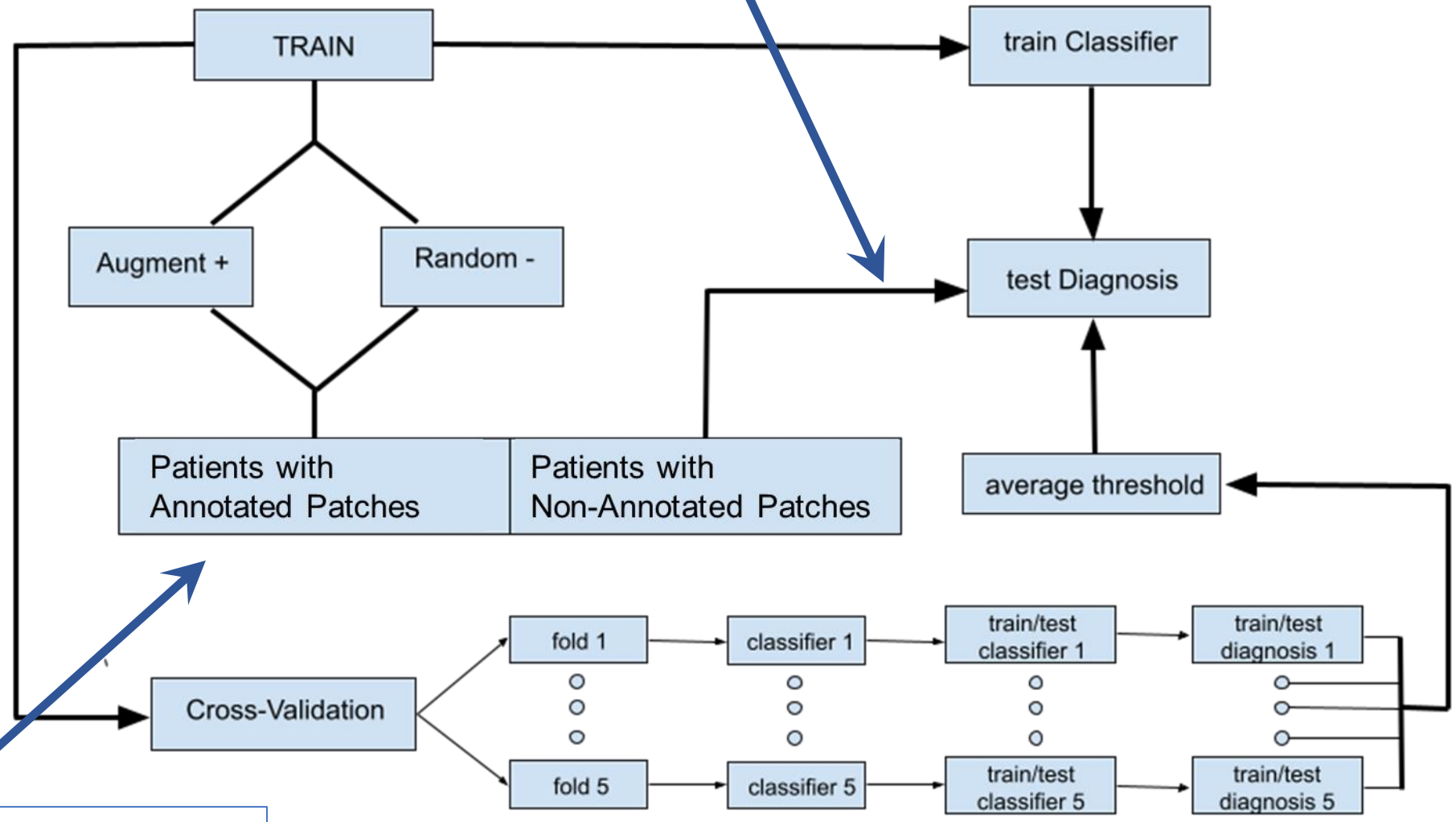
Data preparation:
Extract patches from WSI along sample border

Patch classification:
Identify presence of H.pylori in each patch

Patient diagnosis:
Aggregate patch output for diagnosis

Data Preparation

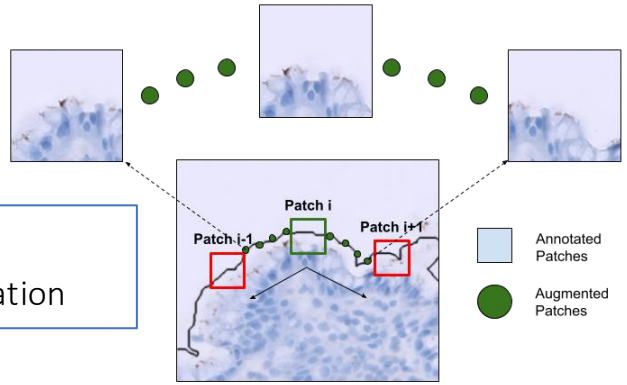
Dataset 2: Independent set of non-annotated patches for assessment of reproducibility



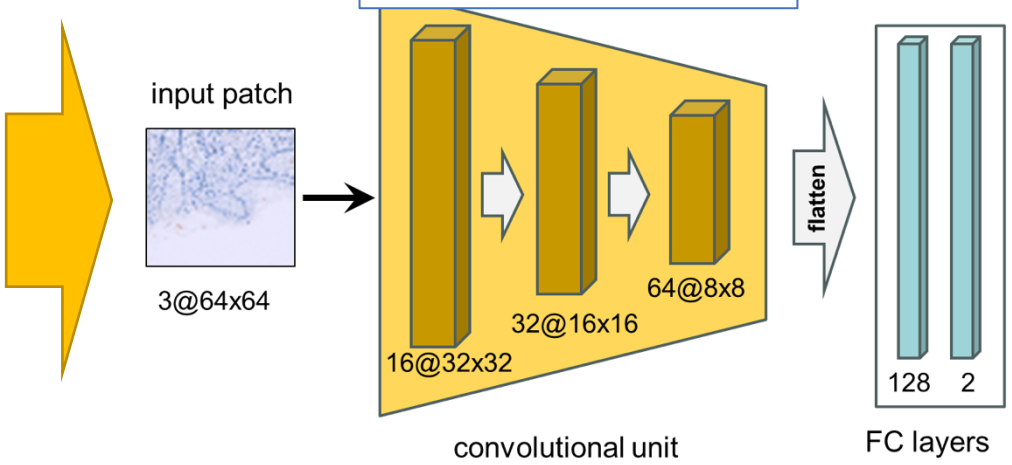
Dataset 1: Patients with annotated patches for cross-validation of models

Patch Classification

Data Augmentation

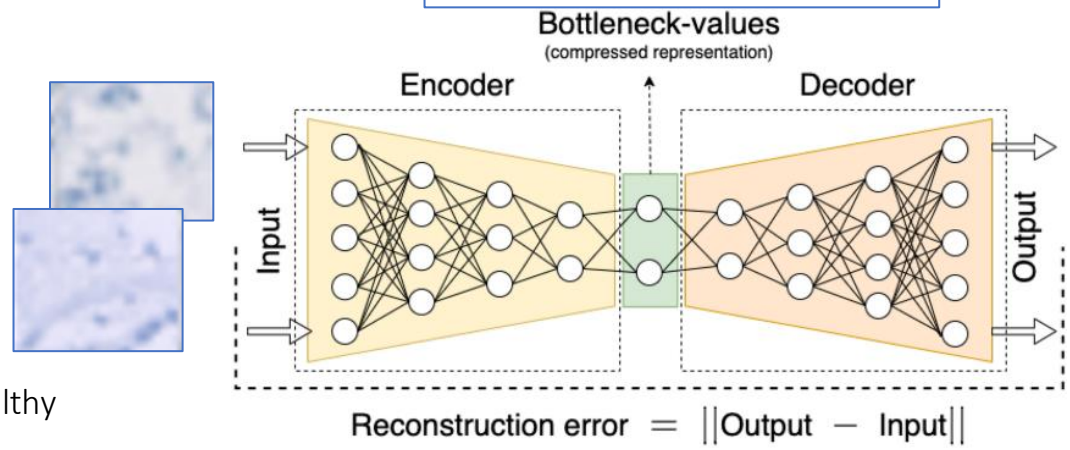


Shallow architecture



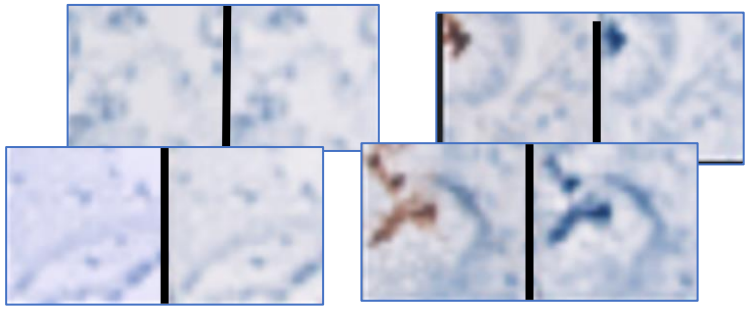
Classification Approach

Self-supervised learning of visual representations.



Healthy

Reconstruction error detects anomalous staining

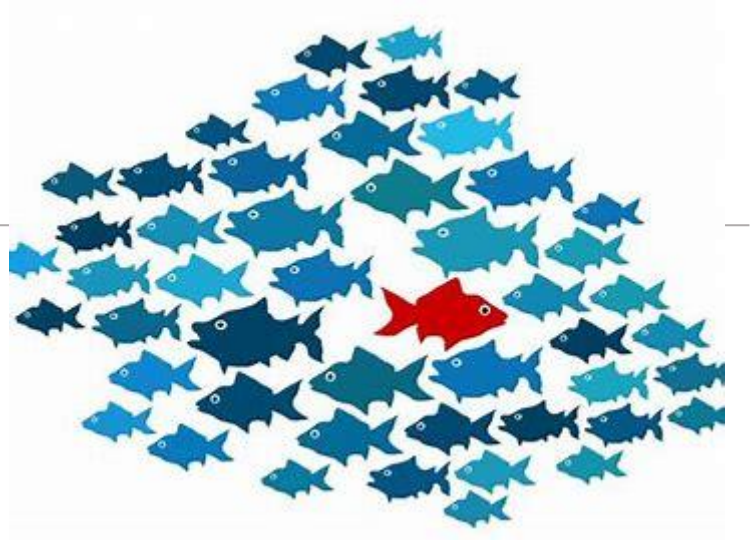


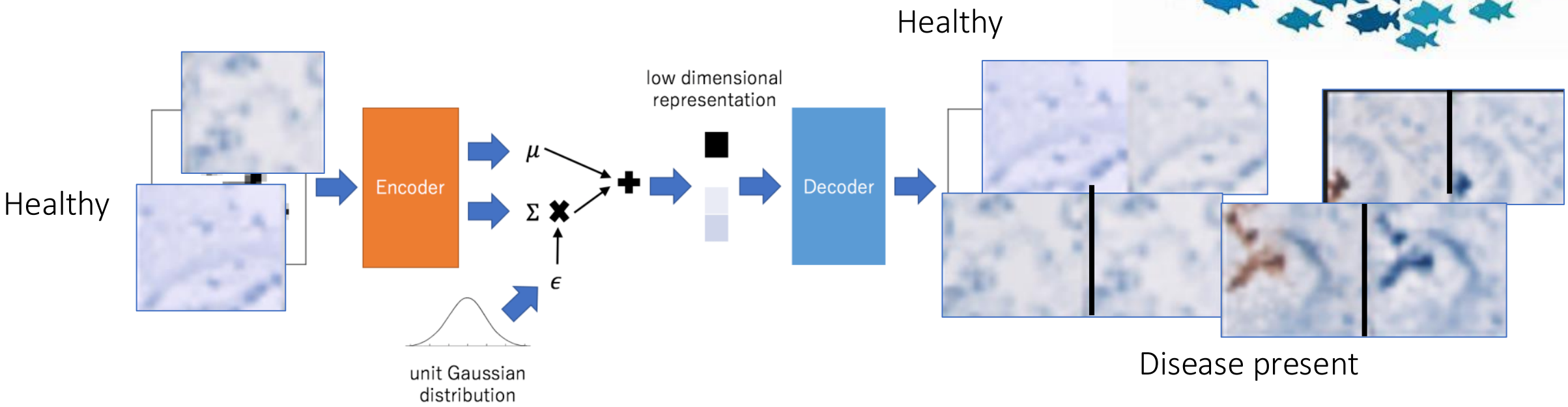
Healthy

Disease present

Anomaly Detector

Anomaly Detection





Learn majority/normality (autoencoders). Use reconstruction error to detect the rare small sample size population (https://arxiv.org/abs/2309.16053)

Use generative models for self-supervised learning of representation spaces

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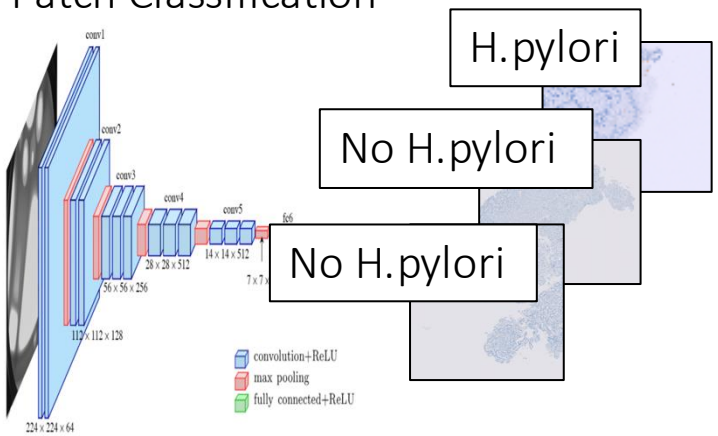
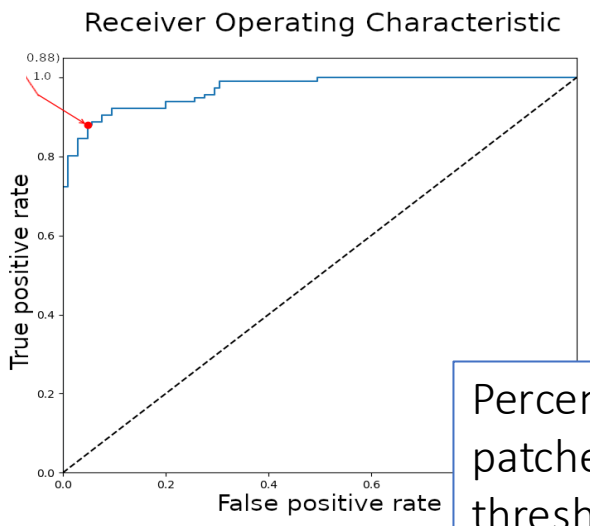
Patient Diagnosis

Adaptive Thresholding

Patch Classification

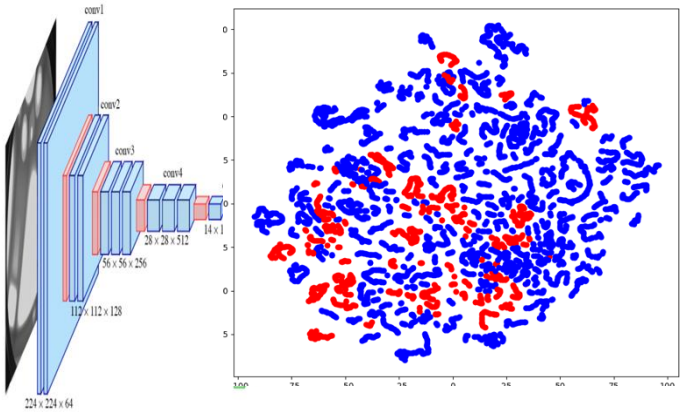


Optimal th

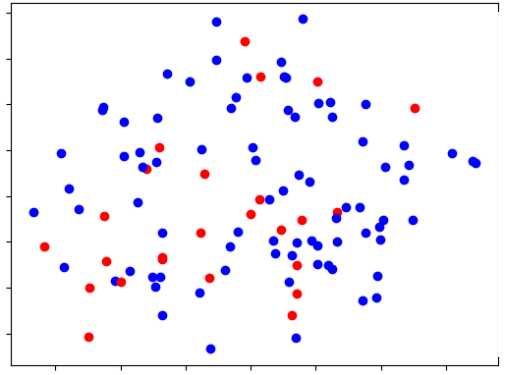


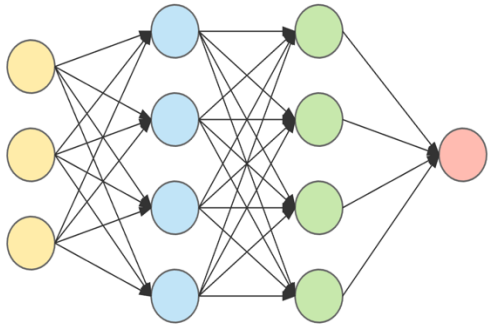
Percentage of positive patches with adaptative thresholding

Attention Mechanism

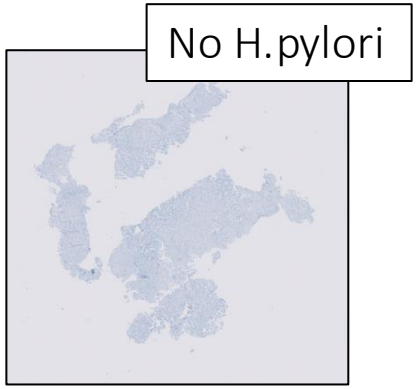


Patch Visual Features





Sample representation space using attention for patient classification



Sample Classification

Specific Objectives

1. Implement two systems:

    1.1 Patch Classification: Anomaly Detector, Classification Approach

    1.2 Patient Diagnosis: Adaptive Thresholding, Attention Mechanism

2. Compare performance:

    2.1 Cross-Validation: patch and patient level

    2.2 Reproducibility: patient level

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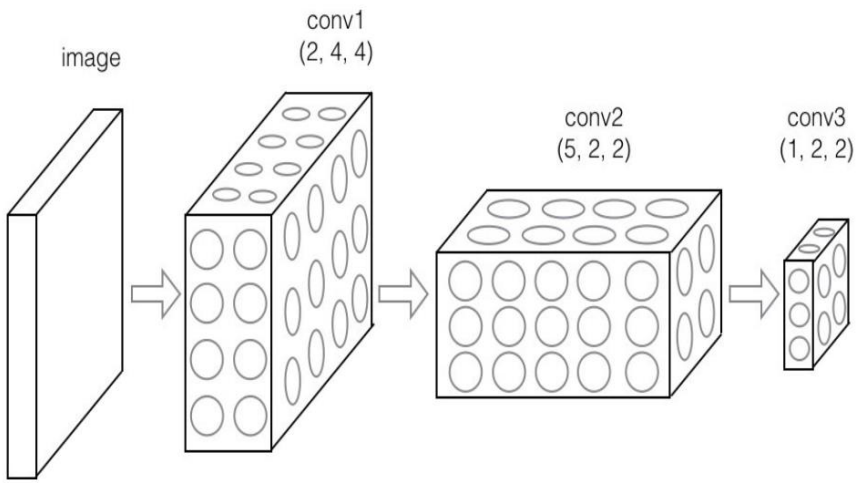
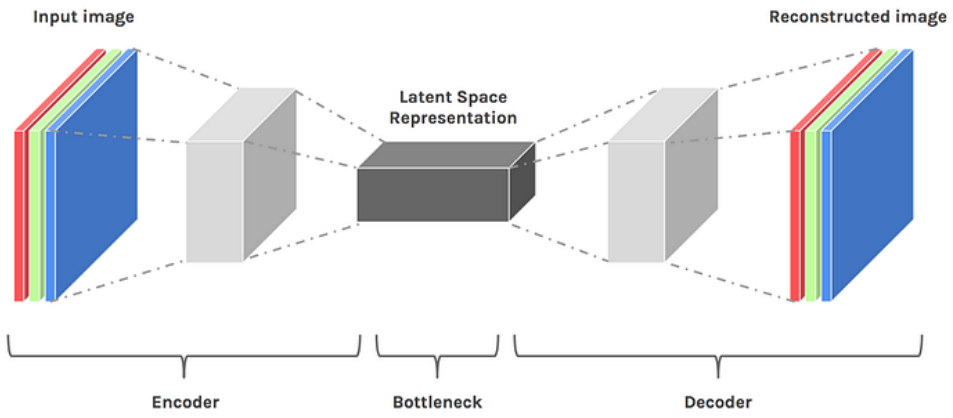
Part 1: using CNN



Part 2: using Autoencoders / Contrastive Learning



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The recipe

- Examine and define the problem
- Explore what you already know
- Determine what you need to learn
- Evaluate possible ways to solve the problem

February	Tuesday 3	Baseline Classifier. Introduction to Neural Networks Materials in virtual campus: Intro MLPs. MPLs for Images.	**15-17**
	Thursday 5	Baseline Classifier. Introduction to CNNs Materials in virtual campus: Intro CNNs. PreTrained CNNs.	**17-19**
	Tuesday 10	Baseline Classifier. Fine tune a pre-trained network to detect Hpylori in annotated windows. Materials in virtual campus: Cropped patches from all WSI. Annotated patches.	**15-17**
	Thursday 12	Baseline Classifier. Patient Diagnosis from Baseline Classifier Materials in virtual campus: Challenge Description. HPyloriDetectionAutoEncoders.pdf, Validation and Verification.	**17-19**
	Tuesday 17	Validation of Models. Materials in virtual campus: Validation and Verification.	**15-17**
	Thursday 19	Project 1: First Defence, Version 1	**17-19**
	Tuesday 24	Design of Generative Architectures (AutoEncoders). Materials in virtual campus: QUIRON Dataset: Cropped Patches from the all WSI, Autoencoders, Challenge Description. HPyloriDetectionAutoEncoders.pdf. Optimization of Hyperparameters.	**15-17**
	Thursday 26	Validation of Models. Experimental Design. Data Split. Metrics Materials in virtual campus: Validation and Verification. Advanced Validation.	**17-19**

March	Tuesday 3	Comparison to Baseline Approach. Experimental Design. Data Split. Metrics. Materials in virtual campus: Validation and Verification. Advanced Validation.	**15-17**
	Thursday 5	Project 1: Final Defence, Version 2	**17-19**

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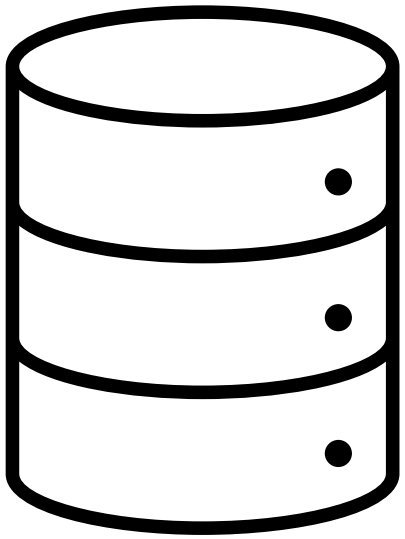
Dataset



Quiron Salut Dataset

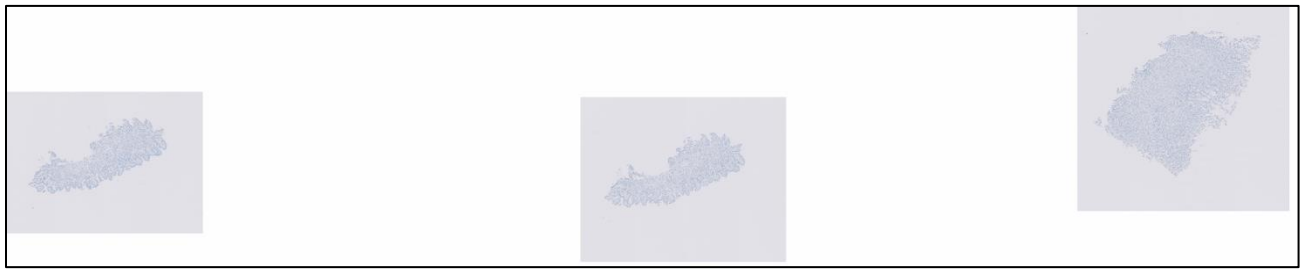
245 gastric biopsies scored according to H.pylori density into

NEGATIVE (117), LOW and HIGH DENSITY

CSV file for the density

CODI,DENSITAT
B22-01,BAIXA
B22-02,BAIXA
B22-03,NEGATIVA
B22-04,NEGATIVA
B22-05,NEGATIVA
B22-06,NEGATIVA
B22-07,NEGATIVA
B22-08,NEGATIVA
B22-09,NEGATIVA
B22-10,NEGATIVA
B22-11,NEGATIVA
B22-12,NEGATIVA

Control to check staining quality



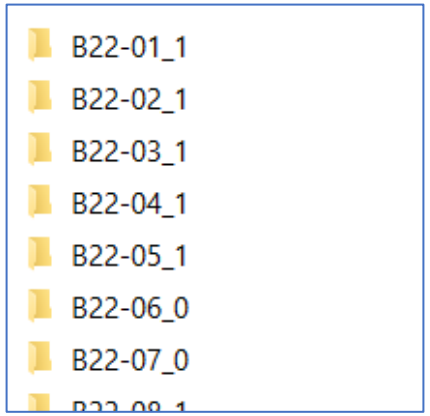
Tiff images with 3 sections

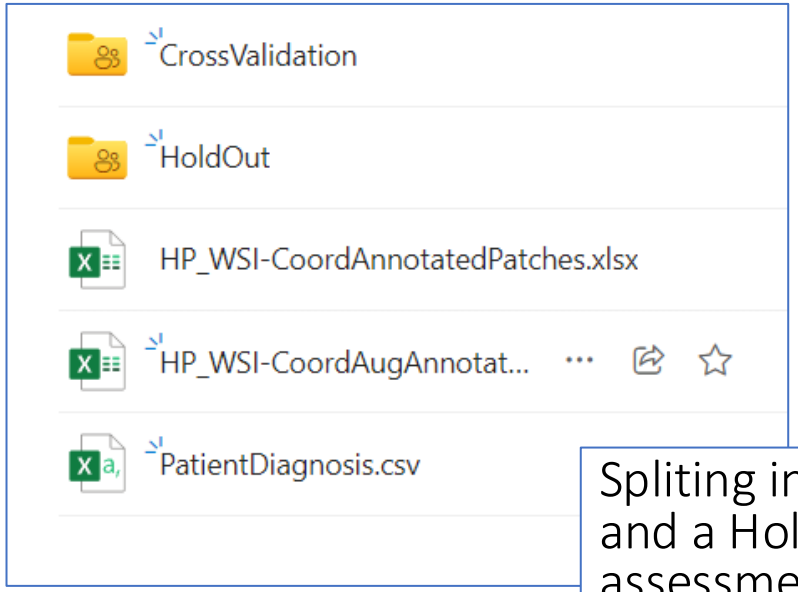
DataSet available through VC and GPU server /fhome/vlia/HelicoDataSet

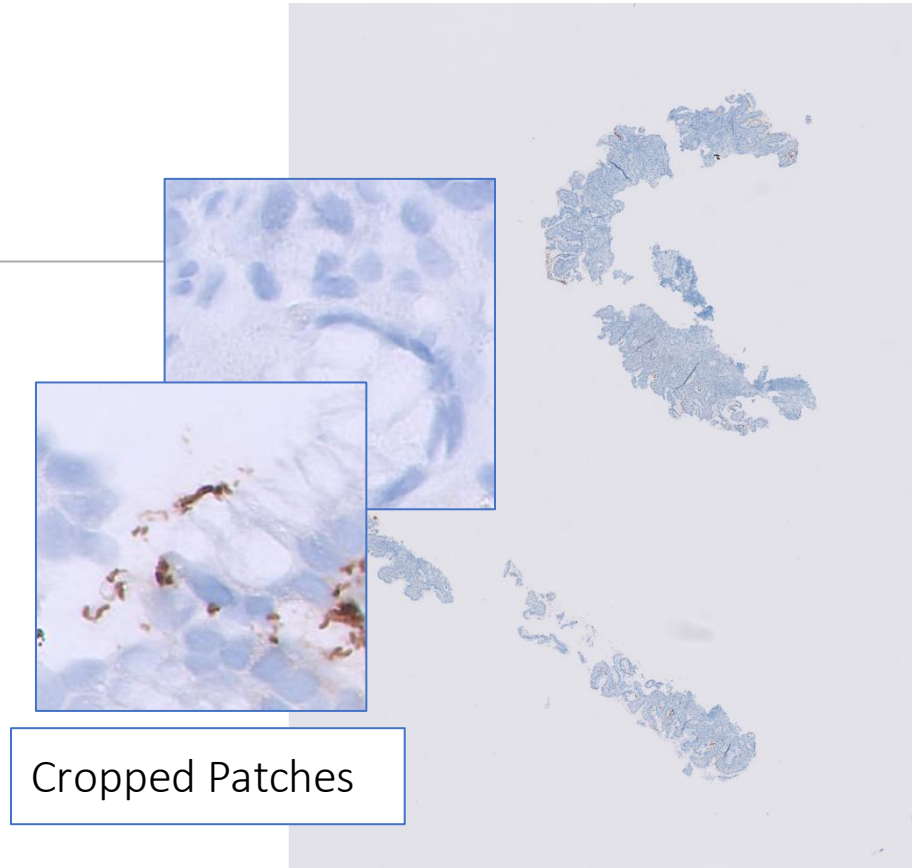
Quiron Salut Dataset

Patches of size 256x256 were cropped along the border of the tissue samples of one of the sections.

Cropped images in PNG format are organized within folders named according to the PatID and tissue section, PatID_Section#





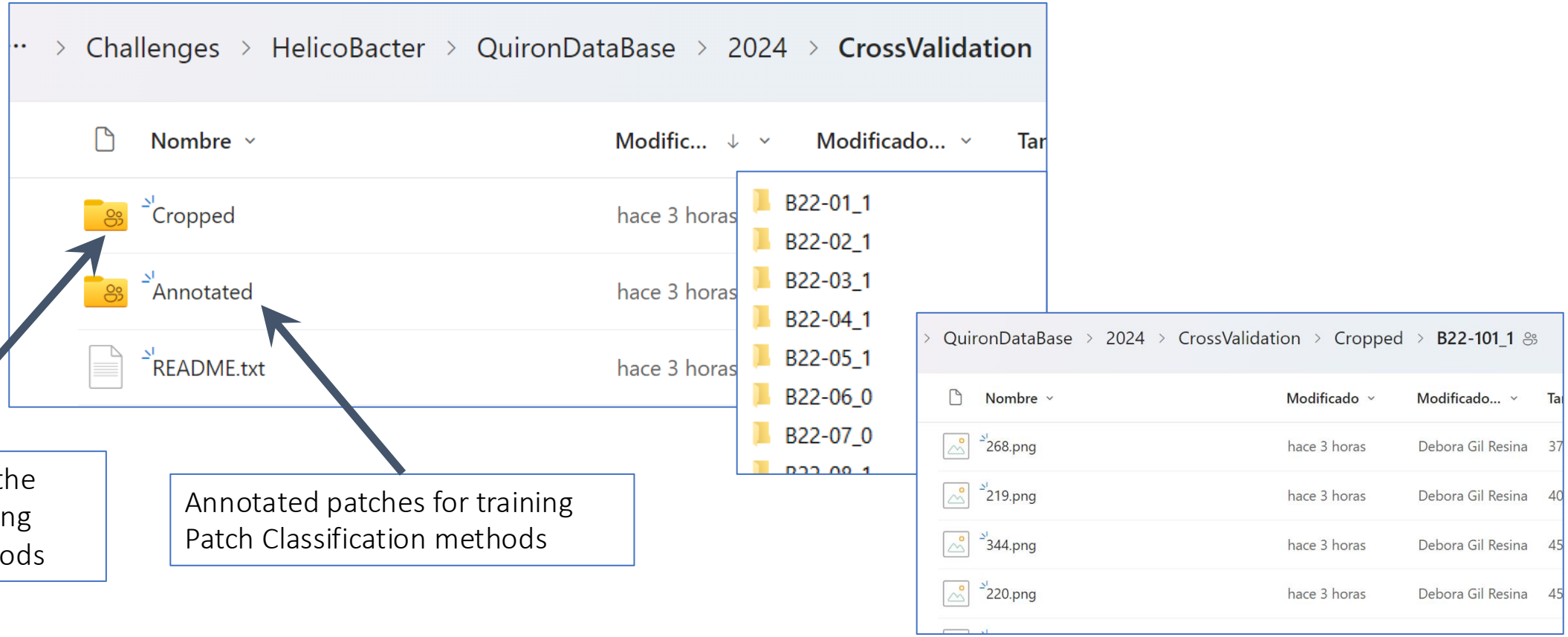


Cropped Patches

Spliting into a CrossValidation set with annotations and a Holdout independent set for reproducibility assessment

Cross-Validation set

From a subset of 123 cases (77 positive ones), an expert pathologist annotated 1211 patches (161 positive). The set was complemented with images from negative cases to balance the set.



Patches cropped from the whole section for training Patient Diagnosis methods

Annotated patches for training Patch Classification methods

Cross-Validation set

HP_WSI-CoordAnnotatedPatches.xlsx

- The coordinates (i,j,w,h) of the annotated patches
- identification and presence of bacteria (-1 No Helico, 1 Helico, 0 Not clear)

This file contains only annotated patches. The remaining ones are negative patches

Pat_ID	WSI_ID	Window_ID	i	j	h	w	Presence
B22-129	0	659	7477	11978	256	256	-1
B22-68	0	131	6597	12009	256	256	-1
B22-68	0	141	5100	10737	256	256	-1
B22-68	0	290	5015	14908	256	256	-1
B22-68	0	298	11626	13928	256	256	-1
B22-68	0	315	9541	11016	256	256	-1
B22-68	0	352	7536	9139	256	256	-1
B22-52	0	760	5664	12310	256	256	-1
B22-124	0	11	47727	30234	256	256	-1
B22-124	0	759	28997	17596	256	256	-1
B22-124	0	762	28701	18129	256	256	-1

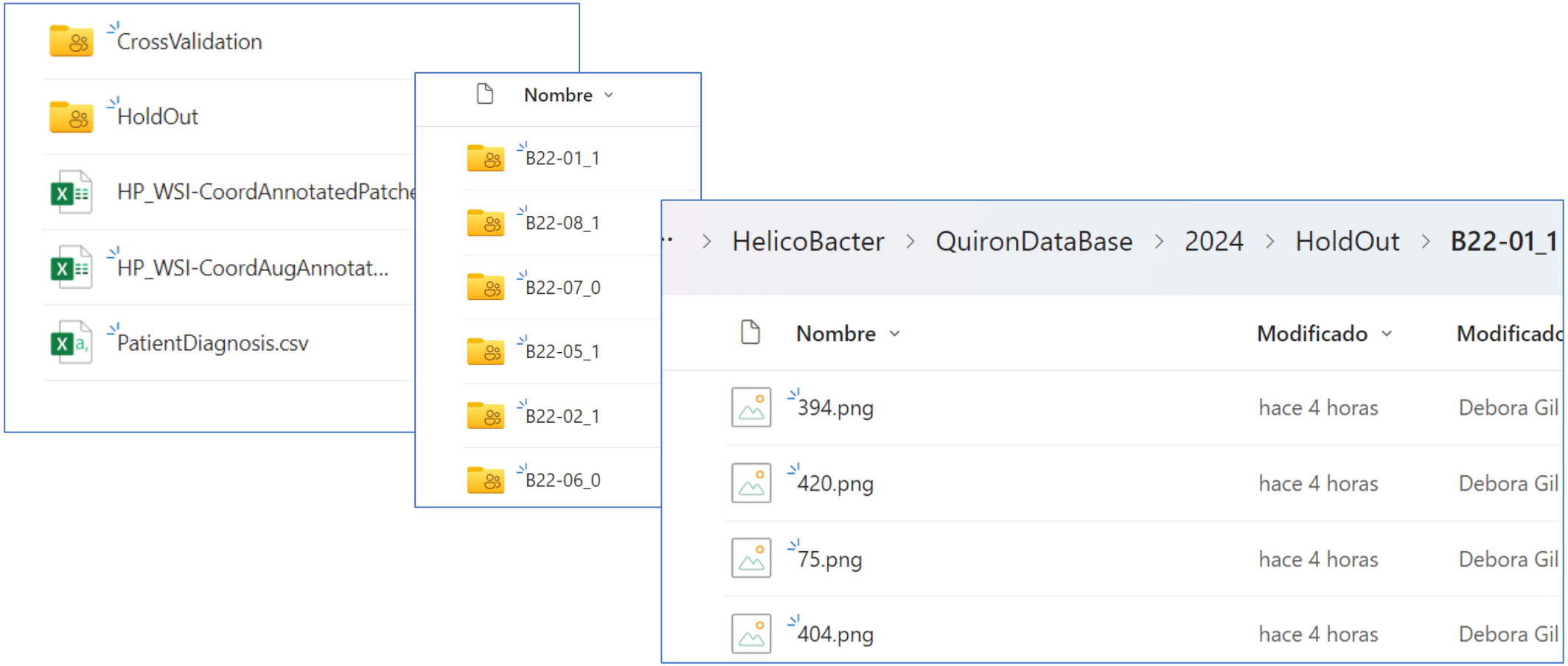
Cross-Validation set

Labeled patches can be explored at
http://ivendis.cvc.uab.es/Digipatics/display.php

User: isbi_user
Password: isbi_pass

Hold-out set

The remaining 116 cases are an independent Hold-out set for verification of the reproducibility of Patient Diagnosis methods



Diagnosis of Helicobacter pylori using AutoEncoders for the Detection of Anomalous Staining Patterns in Immunohistochemically stained Whole Slide

Pau Cano[1,2], Álvaro Caravaca, Debora Gil[1,2], and Eva Musulen[3,4]

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[4] Institut de Recerca contra la Leucèmia Josep Carreras (IJC), 08916 Badalona, Barcelona, Spain emusulen@carrerasresearch.org

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